

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0344

loco 138 kV
August 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0344. The Interconnection Customer proposes a 582.4 MW natural gas combined cycle facility interconnecting at the loco 138 kV substation in the Virginia area. The request was received on June 19, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 144 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$61,712,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sandy Ridge - N Reading 138.0 kV
Contingency: loco - Kean Mtn And Grassy Creek 138.0 kV
Loading: 235.0% (Rating: 242 MVA)
DFAX: 100.0%
Pre-project loading: 5.4%
2. Facility: N Reading Substation - loco Substation 138.0 kV
Contingency: loco - Kean Mtn And Grassy Crk 138.0 kV L
Loading: 234.5% (Rating: 243 MVA)
DFAX: 100.0%
Pre-project loading: 5.0%
3. Facility: loco - Kean Mtn And Grassy Crk 138.0 kV line
Contingency: loco - N Reading 138.0 kV
Loading: 222.7% (Rating: 256 MVA)
DFAX: 100.0%
Pre-project loading: 4.7%
4. Facility: Coal Crk - Sandy Ridge 138.0 kV
Contingency: loco - Kean Mtn And Grassy Crk 138.0 kV Line
Loading: 196.4% (Rating: 286 MVA)
DFAX: 100.0%
Pre-project loading: 7.1%
5. Facility: Permac - Coal Creek 138.0 kV
Contingency: Kean Mtn And Grassy Crk - loco 138.0 kV
Loading: 156.0% (Rating: 191 MVA)
DFAX: 63.0%
Pre-project loading: 36.6%

6. Facility: Permac - Coyote 138.0 kV
Contingency: Kean Mtn And Grassy Crk - loco 138.0 kV Line
Loading: 154.1% (Rating: 188 MVA)
DFAX: 63.0%
Pre-project loading: 40.9%
7. Facility: Garden Creek - Kean Mtn And Grassy Creek 138.0 kV line
Contingency: Kean Mtn And Grassy Creek 345/1kV XFMR
Loading: 112.1% (Rating: 262 MVA)
DFAX: 39.5%
Pre-project loading: 24.3%
8. Facility: Garden Crk - Turkey Pen 138.0 kV L
Contingency: Kean Mtn And Grassy Creek 345 kV/1 kV Transformer
Loading: 110.8% (Rating: 256 MVA)
DFAX: 39.5%
Pre-project loading: 20.8%
9. Facility: Hoart - Haysi Substation 138.0 kV
Contingency: loco - Kean Mtn And Grassy Crk 138.0 kV
Loading: 100.7% (Rating: 206 MVA)
DFAX: 29.8%
Pre-project loading: 16.7%
10. Facility: Kean Mtn And Grassy Creek 345/1 kV Transformer
Contingency: Carlton Hills - Clinch River 345.0 kV
Loading: 100.0% (Rating: 548 MVA)
DFAX: 42.1%
Pre-project loading: 55.3%
11. Facility: Kean Mtn And Grassy Crk 138/1 kV Transformer
Contingency: Carlton Hills - Clinch River 345.0 kV
Loading: 100.0% (Rating: 548 MVA)
DFAX: 42.1%
Pre-project loading: 55.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sandy Ridge - N Reading 138.0 kV	Rebuild line with higher-capacity conductor	\$2,135,000	100% (\$2,135,000)
North Reading - loco 138.0 kV	Rebuild line with higher-capacity conductor	\$2,573,000	100% (\$2,573,000)
Kean Mtn And Grassy Crk - loco 138.0 kV	Rebuild line with higher-capacity conductor	\$5,646,000	100% (\$5,646,000)
Coal Crk - Sandy Ridge 138.0 kV	Rebuild line with higher-capacity conductor	\$3,621,000	100% (\$3,621,000)
Permac - Coal Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$4,456,000	100% (\$4,456,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Permac - Coyote 138.0 kV	Rebuild line with higher-capacity conductor	\$21,432,000	100% (\$21,432,000)
Garden Creek - Kean Mtn And Grassy Creek 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,066,000	100% (\$4,066,000)
Turkey Pen - Garden Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$4,982,000	100% (\$4,982,000)
Haysi - Hobart 138.0 kV	Reconductor line	\$5,177,000	54% (\$2,813,000)
Kean Mtn And Grassy Creek 345/1 kV Transformer	Replace transformer with higher-rated unit	\$12,519,000	50% (\$6,242,000)
Kean Mtn And Grassy Crk 138/1 kV Transformer	Replace transformer with higher-rated unit	\$7,512,000	50% (\$3,746,000)
	Total allocated to project		\$61,712,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.